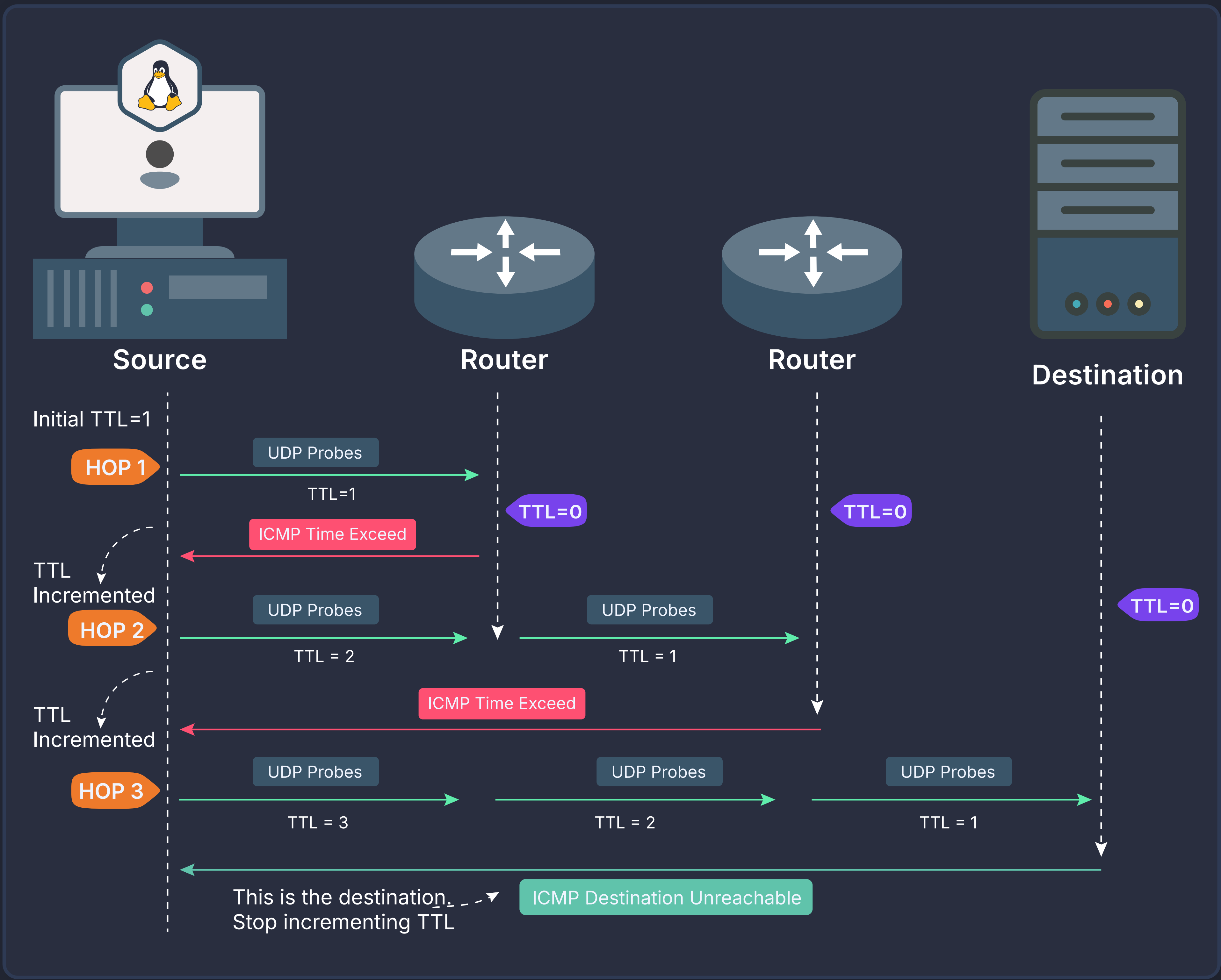
How Traceroute Works?



LINUX

```
$ traceroute [options] host
$ traceroute --help

-4       Use IPv4
-6       User IPv6
-d       Enable Socket Level Debuging
-f       Custom TTL Value instead of 1
-I       Use ICMP ECHO for traceroute
-T       Use TCP SYN for tracerouting
-m       Custom MAX number of hops
-s       Use custom src address for
         outgoing packets
-p       Set Destination UDP port to
         use instead of default(33434)
```

WINDOWS

```
$ tracert [options] host
$ tracert --help

-4       Use IPv4
-6       User IPv6
-d       Do not resolve addresses to
         hostnames.
-h       Custom MAX number of hops
-S       Custom source address to use
         (works with IPv6 only)
```

Key Concepts

Concept	Explanation
Time To Live (TTL)	A field in the IP header that limits a packet's lifespan. TTL is decremented by 1 at each router. When it reaches 0, the router discards the packet and send ICMP Time Exceeded response.
ICMP Time Exceeded Message	Sent by a router when a packet's TTL reaches 0. Informs the sender of the packet's expiration, revealing the router's IP address.
Probes``	For each TTL value, traceroute sends 3 probe packets by default (configurable with -q on Linux) to measure response-time consistency (RTT - Round Trip Time).

Protocol Variation

Protcol	Explanation
UDP-Based Tracerouting	Sends UDP packets to high-numbered, unused ports. Routers return ICMP Time Exceeded messages to reveal hops. Default method on Linux and macOS. May be blocked or filtered by some firewalls.
ICMP-Based Tracerouting	Uses ICMP Echo Request packets (like ping). Routers send ICMP Time Exceeded when TTL expires. Default on Windows. Some networks rate-limit or deprioritize ICMP.
TCP-Based Tracerouting	Sends TCP SYN packets to a chosen port (e.g., 80/443). Helps bypass firewalls that block UDP/ICMP but allow TCP. Final host replies with SYN/ACK (open) or RST (closed). Useful for checking paths to specific services.